WORKSHEET 9.1

How the Exam Is Built and Scored

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- **Section I**: 29 multiple-choice questions in 62 minutes without a calculator, then 13 in 38 minutes with one.
- **Section II**: 2 free-response questions in 30 minutes with a calculator, then 4 in 60 minutes without one.
- Each multiple-choice question is worth 1 point and each free-response question 9. The two sections count equally. A wrong multiple-choice answer costs nothing.
- On free response, the set-up earns points. Write the integral, equation or derivative before its value.

Worked examples

Example 1 · Pacing

In the calculator multiple-choice part you have answered 9 of the 13 questions and 10 minutes remain. Are you on pace?

Solution. 4 questions in 10 minutes is 2.5 minutes each; the average for the part is about 2.9.

Slightly behind. Answer what you can quickly and leave no question blank.

Example 2 · Combining the sections

A student earns 34 of 42 multiple-choice points and 27 of 54 free-response points. What fraction of the total is this?

Solution. $\dfrac{1}{2}\left(\dfrac{34}{42}+\dfrac{27}{54}\right)$.

About 0.655, or 65.5%.

Example 3 · Decimal answers

The value of an integral is $3.14159\ldots$ Which of 3.1, 3.141 and 3.142 would be accepted?

Solution. A decimal answer must be correct to three places after the point, rounded or truncated.

3.141 **and** 3.142**, but not** 3.1**.**

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Practice

CORE

1. How many free-response questions are there, and on how many of them is a calculator used?

2. How long is Section I in total?

3. What is the greatest number of points available in Section II?

4. What is the average time per question in the no-calculator multiple-choice part?

EXAM LEVEL

5. A student earns 25 of 42 multiple-choice points and 40 of 54 free-response points. What fraction of the total is this?

6. A student expects 36 of 54 on the free response and wants at least 60% overall. How many multiple-choice questions must be correct?

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7. With 20 minutes left in the no-calculator free-response part, two questions are untouched. Describe a sensible plan.

8. In part (b) you need your answer to part (a), which you suspect is wrong and cannot fix. What should you do?

Question 9 · Free-response practice

No calculator. Write each answer as you would in the exam: set-up first, then the value, then the reason. Let $f(x) = x^3 - 3x^2$.

(a) Find the average value of f on $[0, 2]$.

(b) Write an equation of the line tangent to the graph of f at $x = 1$.

(c) Show that f has a relative minimum at $x = 2$. Justify your answer.

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Worked solutions

1. 6 questions; a calculator is used on 2.
2. $62 + 38 = 100$ minutes.
3. $6 \times 9 = 54$.
4. $\dfrac{62}{29} \approx 2.1$ minutes.
5. $\dfrac{1}{2}\left(\dfrac{25}{42} + \dfrac{40}{54}\right) \approx 0.668$.
6. $\dfrac{1}{2}\left(\dfrac{x}{42} + \dfrac{36}{54}\right) \geq 0.6$ gives $x \geq 22.4$: at least 23.
7. About 10 minutes each. Read every part of both, write the set-ups you can, and attempt the parts that stand alone first.
8. Use your own value and carry on. Work that follows correctly from an earlier error can still earn full credit.
9. (a) $\dfrac{1}{2}\displaystyle\int_0^2 (x^3 - 3x^2)\,dx = \frac{1}{2}\left[\tfrac{x^4}{4} - x^3\right]_0^2 = \frac{1}{2}(-4) = -2$.
   (b) $f(1) = -2$ and $f'(1) = 3 - 6 = -3$: $y + 2 = -3(x - 1)$.
   (c) $f'(x) = 3x(x - 2)$, so $f'(2) = 0$, and f' changes from negative to positive at $x = 2$. Therefore f has a relative minimum there.

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